Privacy-Preserving Smart Classroom Occupancy Detection and Energy Management

USER STUDY / FIELD STUDY — RESPONSE ANALYSIS

Total responses	Participant mix	Target
36	Students, teaching staff, administrators/facilities staff, and other respondents	At least 30 responses where applicable

Purpose. This report summarizes the 36 responses in the uploaded spreadsheet. The questionnaire explores classroom attendance, perceived energy waste, usefulness of occupancy monitoring, privacy expectations, and dashboard priorities.

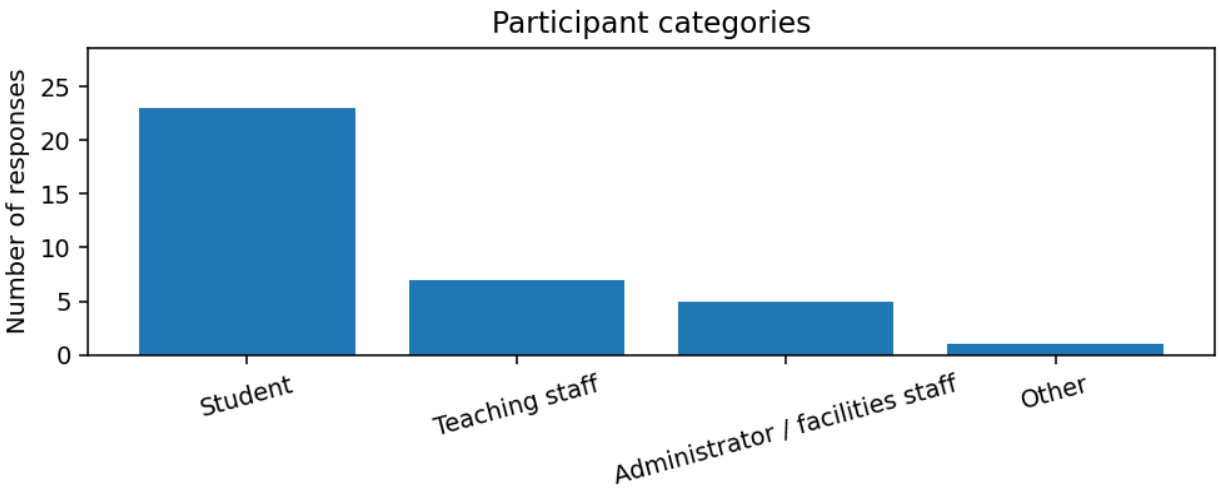
Data note. Results are calculated from the supplied responses. They represent respondents' opinions, not measured technical accuracy or verified energy savings.

1. Executive Summary

The survey received 36 responses, exceeding the target of 30. Most respondents were students (23; 63.9%). Fourteen respondents (38.9%) reported often or very often noticing lights or fans running in empty classrooms or rows. Storage or sharing of camera footage was the most frequently selected camera-related concern (14; 38.9%). Twenty respondents (55.6%) said they would probably or definitely support the system if it works reliably and protects privacy, while 11 (30.6%) were probably or definitely not willing and 5 (13.9%) were unsure.

The results indicate the importance of privacy safeguards, reliable entry/exit counting, occupancy-only dashboard displays, and energy-consumption reporting.

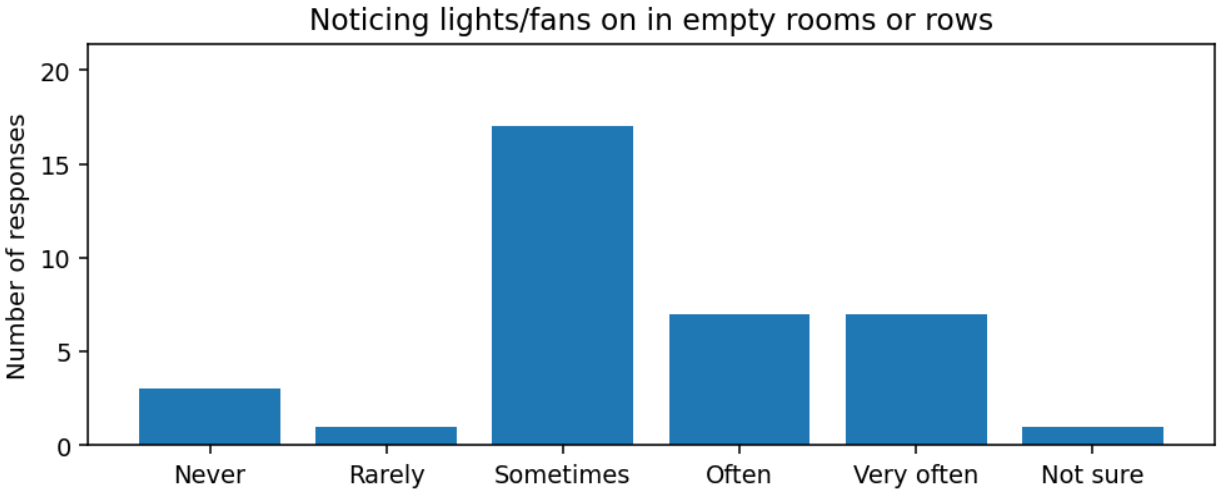
2. Participant Profile



Participant group	Count	Percentage
Student	23	63.9%
Teaching staff	7	19.4%
Administrator / facilities staff	5	13.9%
Other	1	2.8%
Total	**36**	**100.0%**

Classroom-use frequency: 13 respondents reported daily use, 14 several times a week, 6 weekly, and 3 less often.

3. Perceived Energy Waste



14 of 36 respondents (38.9%) selected Often or Very often. Seventeen (47.2%) selected Sometimes; three (8.3%) Never; one (2.8%) Rarely; and one (2.8%) Not sure.

This indicates that respondents perceive unnecessary lighting/fan use as an issue, although the survey does not quantify actual electricity consumption.

4. Current Classroom Problems

Problem selected	Selections
Time-consuming attendance	8
Lights or fans left on unnecessarily	8
Difficulty knowing classroom occupancy	7
Lack of energy-consumption information	5
Attendance recording errors	3
None of these	2
Privacy concerns about monitoring	2
Other	1

Time-consuming attendance and lights/fans left on unnecessarily were the most frequently selected issues (8 selections each), followed by difficulty knowing classroom occupancy (7). Multiple selections were permitted.

5. Ratings of Proposed Features

Feature	Valid answers	Mean (1–5)	Ratings 4–5
Automatic entry/exit detection	36	3.44	18 (50.0%)
Occupancy-based light/fan switching	36	3.42	17 (47.2%)
Row-wise lighting/fan control	35	3.43	18 (51.4%)

Ratings used a 1–5 scale. Means were approximately 3.4, indicating moderate perceived usefulness/importance. Half rated entry/exit detection 4 or 5; 47.2% rated occupancy-based switching 4 or 5; and 51.4% of 35 valid responses rated row-wise control 4 or 5.

6. Camera Privacy and Safeguards

Camera-related concern	Selections
Storage or sharing of footage	14
Incorrect entry/exit counts	9
No concerns	5
Face recognition or identification	4
Continuous video recording	2
Other	2

Preferred safeguards

Preferred safeguard	Selections
Display only occupancy counts and events	9
Do not save video footage	7
Restrict dashboard access	7
Process video locally on the Raspberry Pi	7
Do not use face recognition	4
Other	1
Clearly explain data retention	1

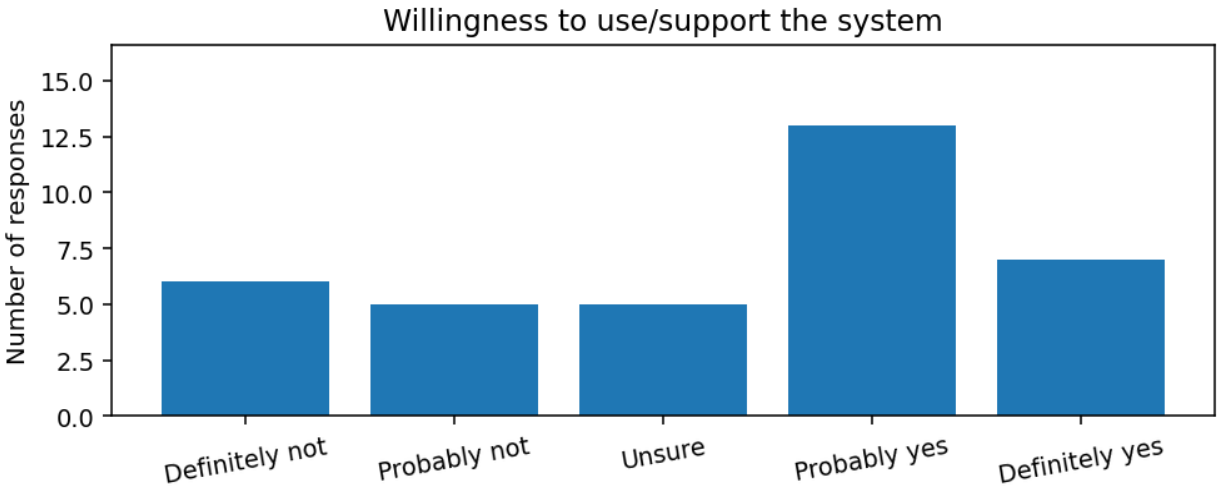
Storage/sharing of footage was the leading concern. The most selected safeguard was displaying only occupancy counts and events. These findings support avoiding face recognition, minimizing stored footage, processing video locally where feasible, and restricting dashboard access.

7. Dashboard Priorities

Dashboard feature	Selections
Energy-consumption reports	19
Attendance summary	18
Current classroom occupancy	17
Light and fan status	17
Row-wise occupancy status	15
Temperature monitoring	12
Noise alerts	12
Daily or weekly reports	11

Energy-consumption reports were selected most often (19), followed by attendance summary (18), current occupancy (17), and light/fan status (17). Some responses selected more than the requested three features, so counts represent all recorded selections and should be interpreted cautiously.

8. Willingness to Use or Support the System



Positive support: 20/36 (55.6%). Negative responses: 11/36 (30.6%). Unsure: 5/36 (13.9%).

A modest majority expressed willingness to support the system under the stated reliability and privacy conditions. The negative responses show that privacy and reliability should be addressed clearly.

9. Open-Ended Feedback

Two respondents provided an answer in the open-ended improvement field; 34 responses were blank. The entries were “Extremely Good For Smart Classrooms” and “E”. Since there are very few comments and one is unclear, no strong theme can be concluded from this question.

10. Recommendations for the Project

• Prioritize an energy dashboard with energy reports, attendance summaries, live occupancy, and appliance status.

• Use privacy-by-design: avoid face recognition, process video locally where feasible, minimize retention, and show counts/events instead of unnecessary footage.

• Test entry/exit counting under realistic conditions and measure accuracy before using counts for attendance decisions.

• Clearly explain what is captured, what is stored, who can access it, and how long data is retained.

• Evaluate energy savings using actual power measurements and before/after comparisons; survey responses alone cannot establish savings.

• Recognize that students make up nearly two-thirds of the sample, so staff/facilities perspectives are less represented.

11. Conclusion

The 36 responses indicate perceived concerns about unnecessary energy use, attendance efficiency, and classroom occupancy visibility. Respondents also identified privacy concerns about camera footage. The project should combine reliable person counting with strong privacy safeguards and a dashboard emphasizing energy reports, attendance summaries, occupancy, and appliance status. Technical testing and actual energy measurements are needed to verify system effectiveness.

Data source: User-uploaded workbook “project phase.xlsx”, sheet “Form Responses 1”. All counts and percentages were calculated from that sheet.